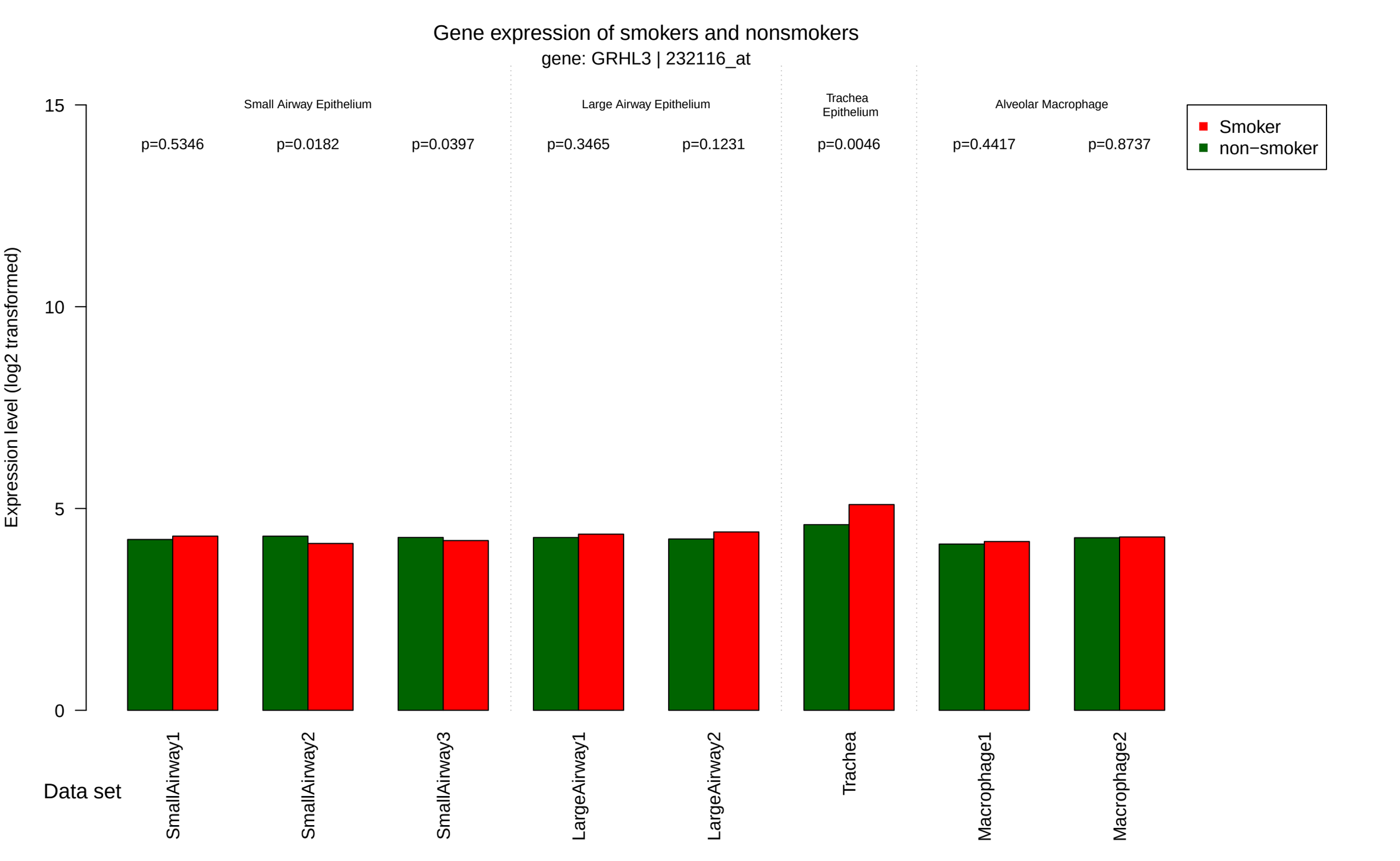
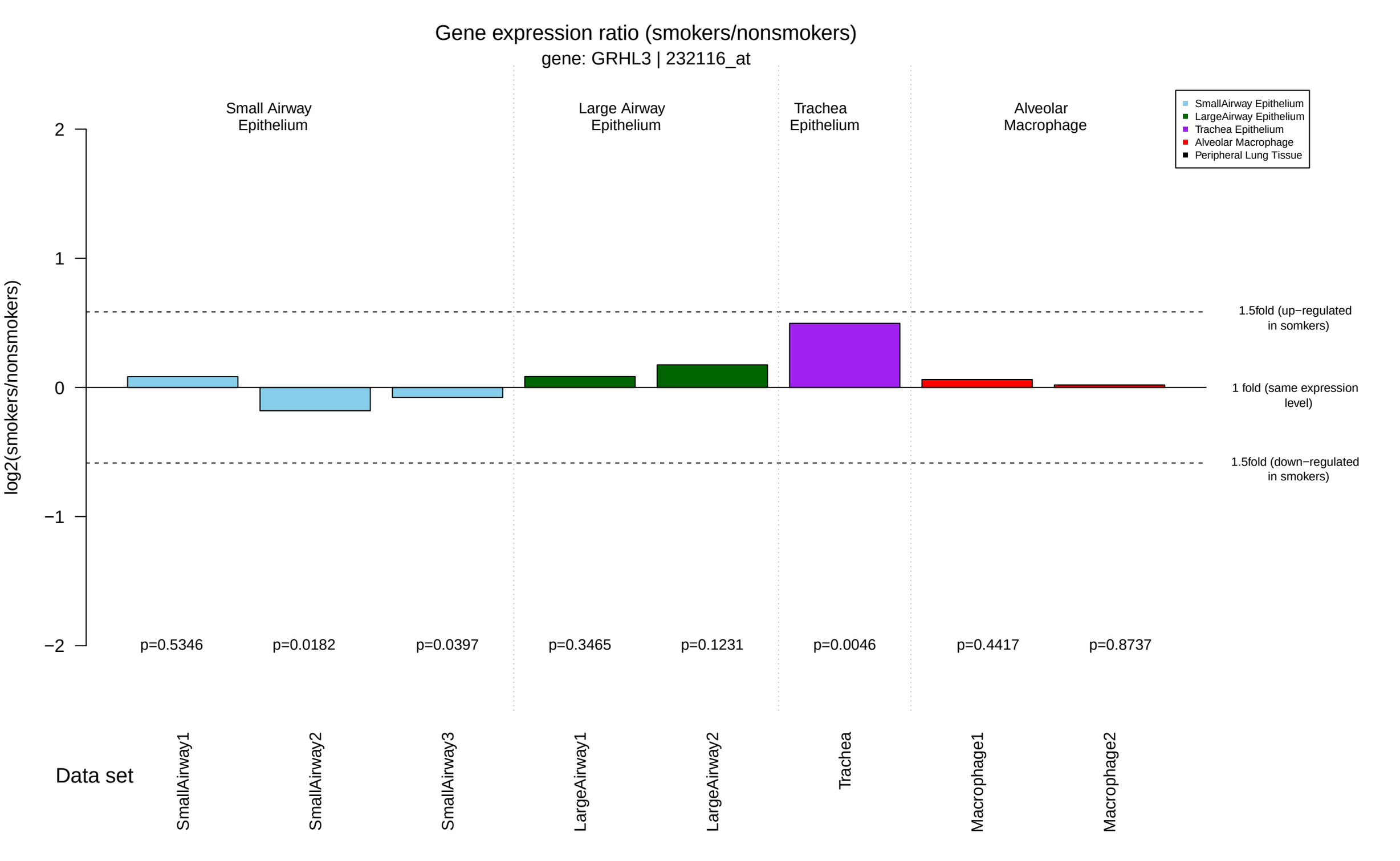
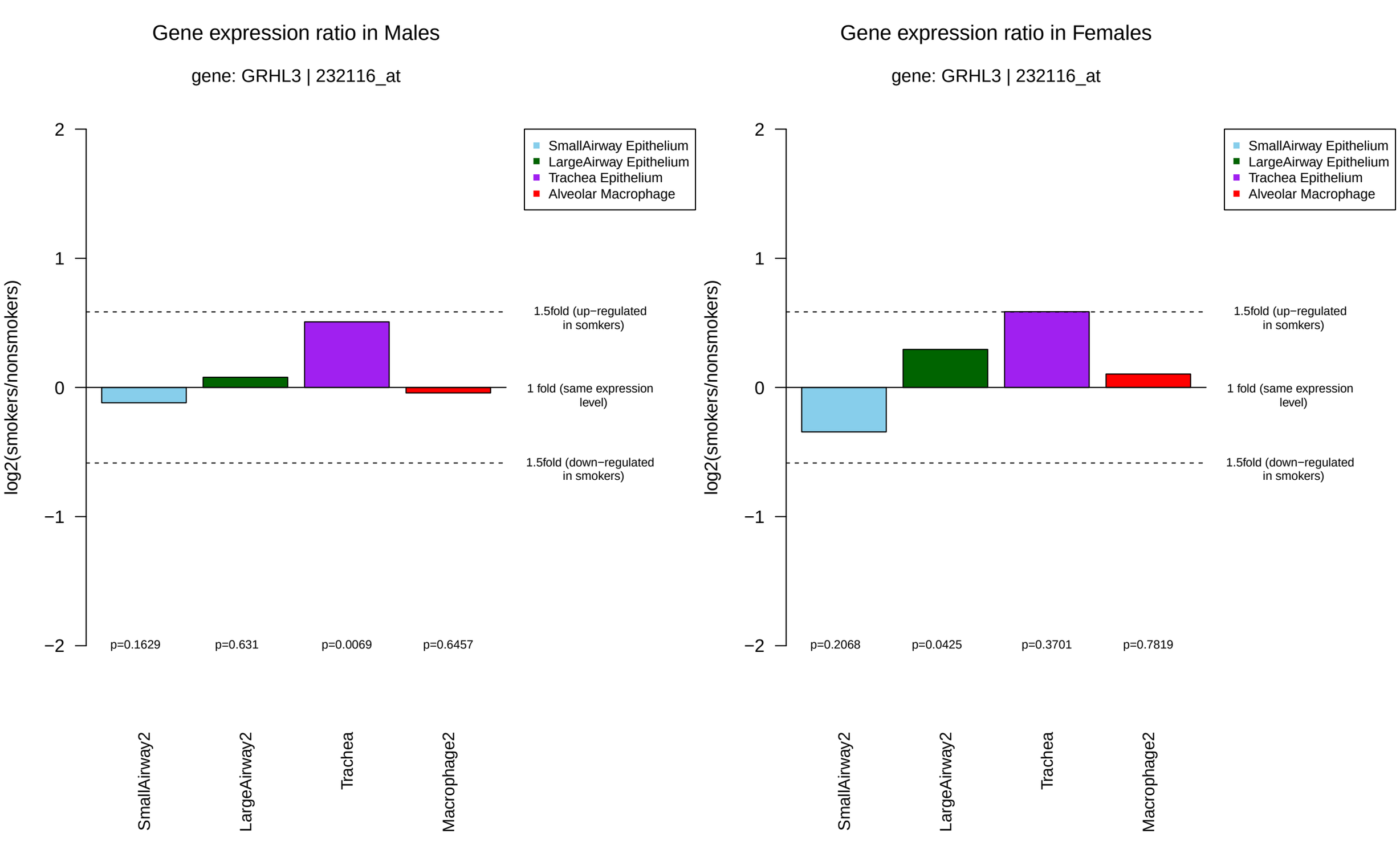
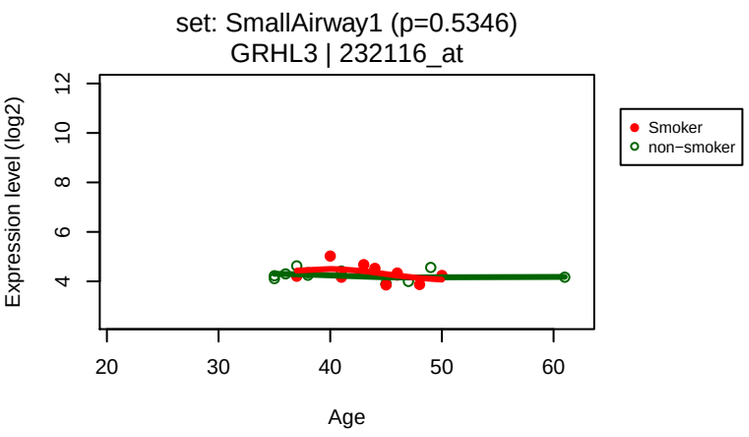
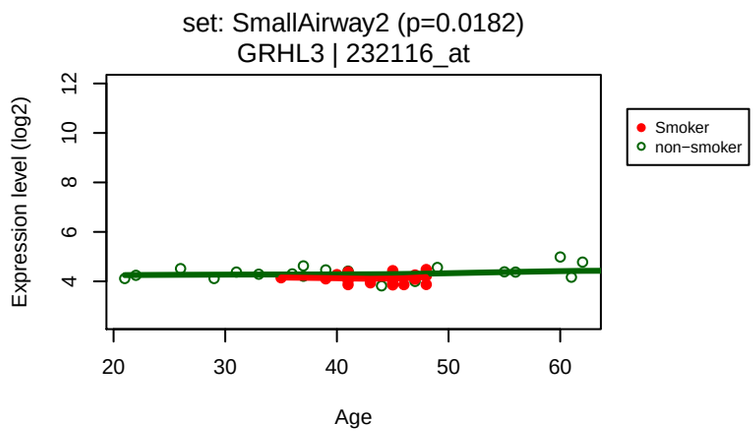
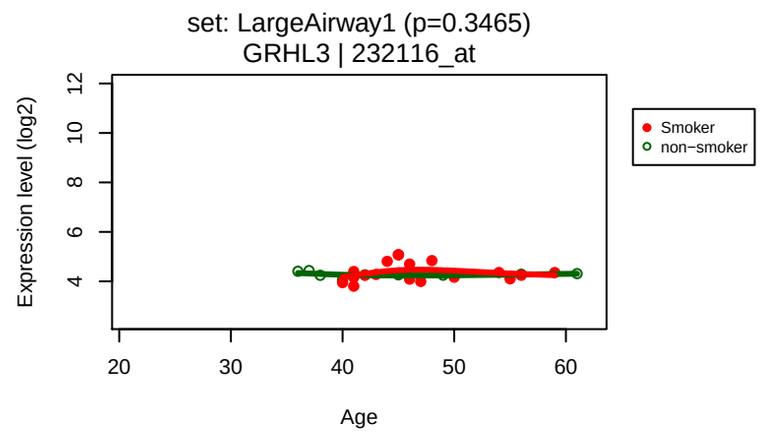

Gene expression of smokers and nonsmokers
gene: GRHL3 | 232116_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5346
p=0.0182
p=0.0397
p=0.3465
p=0.1231
p=0.0046
p=0.4417
p=0.8737
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

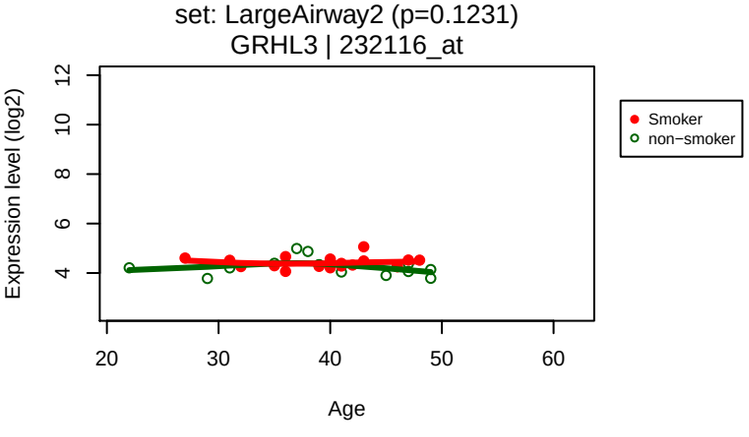

Gene expression ratio (smokers/nonsmokers)
gene: GRHL3 | 232116_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5346
p=0.0182
p=0.0397
p=0.3465
p=0.1231
p=0.0046
p=0.4417
p=0.8737
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

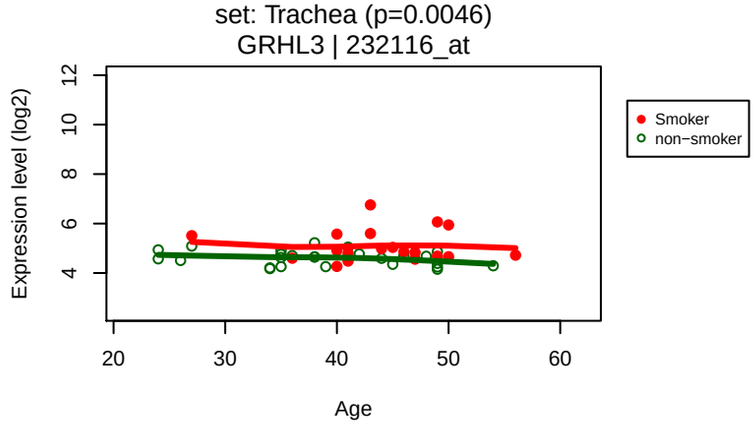

Gene expression ratio in Males
gene: GRHL3 | 232116_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1629
p=0.631
p=0.0069
p=0.6457
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRHL3 | 232116_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2068
p=0.0425
p=0.3701
p=0.7819
SmallAirway2
LargeAirway2
Trachea
Macrophage2

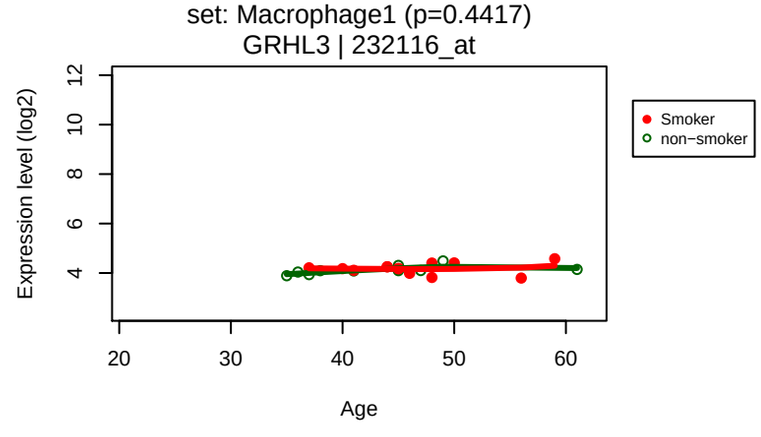

set: SmallAirway1 (p=0.5346)
GRHL3 | 232116_at
Expression level (log2)
Age
Smoker
non−smoker

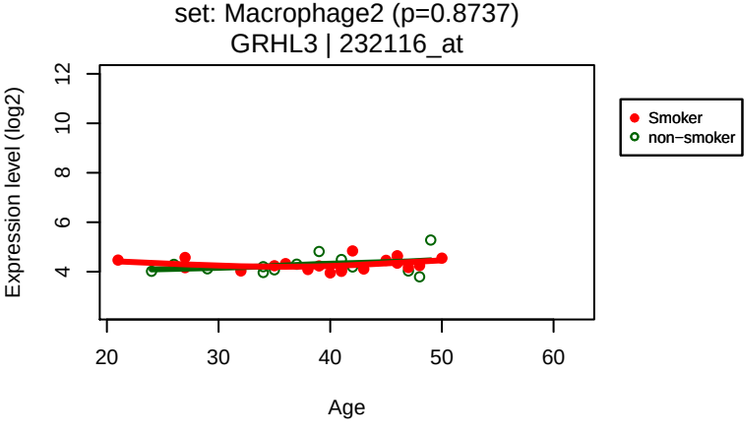

set: SmallAirway2 (p=0.0182)
GRHL3 | 232116_at
Expression level (log2)
Age
Smoker
non−smoker

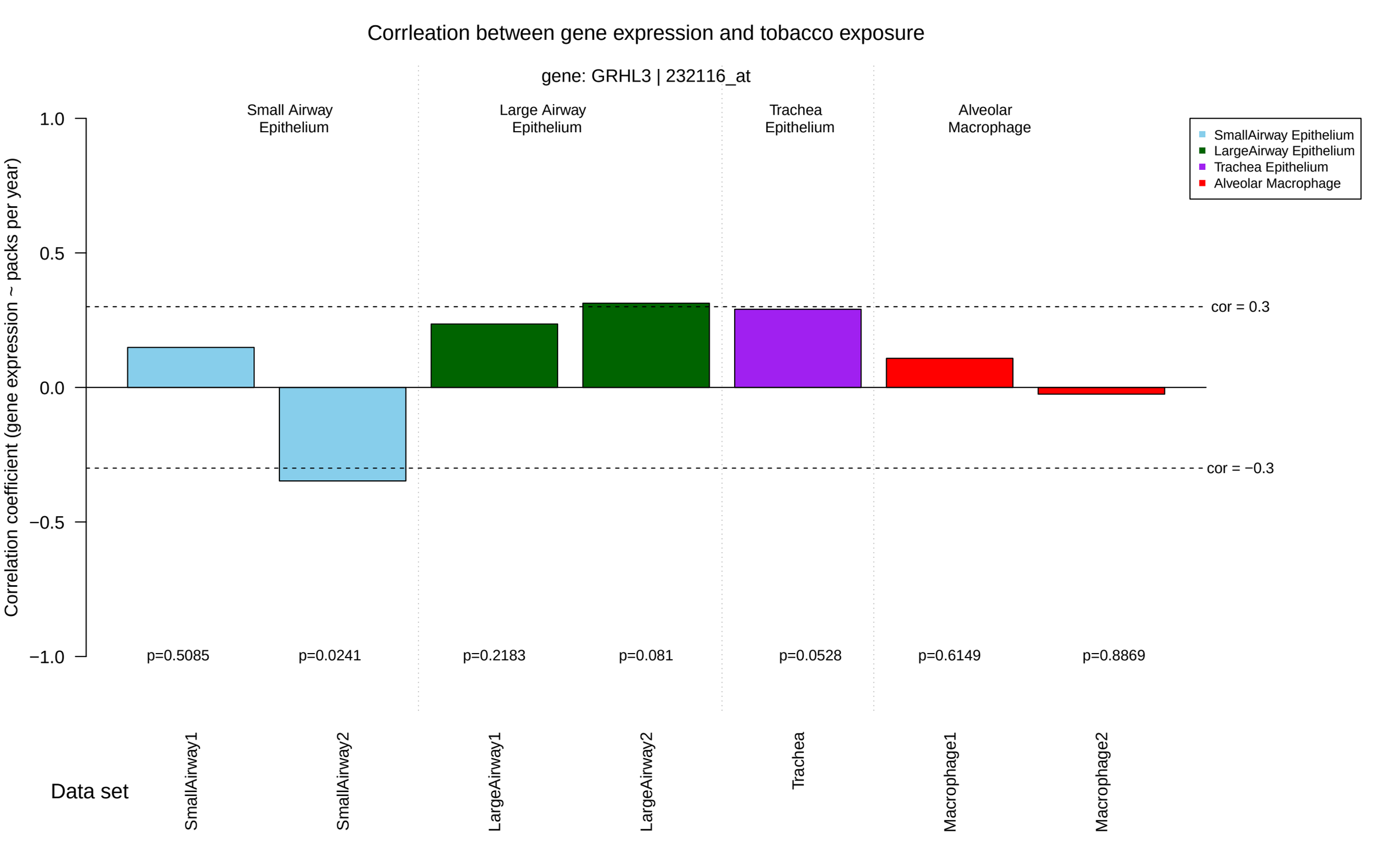

set: LargeAirway1 (p=0.3465)
GRHL3 | 232116_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1231)
GRHL3 | 232116_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0046)
GRHL3 | 232116_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4417)
GRHL3 | 232116_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8737)
GRHL3 | 232116_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRHL3 | 232116_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5085
p=0.0241
p=0.2183
p=0.081
p=0.0528
p=0.6149
p=0.8869
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2